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Front-End Web Development

- **Web:** It refers to websites, web pages, or anything that works over the internet.
- **Development:** It refers to building the application from scratch.

Presented by:

Ujjwal Pudasaini

Front-End Web Development Instructor,

Broadway Infosys

Front-End Technologies



HTML5



CSS3



JavaScript



AngularJS



ReactJS



Introduction to HTML

HyperText Markup Language is the standard document format for defining browser layouts.

- It acts strictly as a structural markup format, not a programming language.
- Instructs the browser exactly how text, images, and tools are structured.
- Saves layouts as structured node hierarchies known as DOM nodes.

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```



Anatomy of an HTML Element

- An HTML element is defined by a start tag, some content, and an end tag.

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple HTML Page</title>
</head>
<body>

  <h1>Welcome to My Website</h1>

  <p>This is a simple HTML .</p>

  <a href="https://www.abc.com">Visit ABC</a>

  <ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
  </ul>

</body>
</html>
```

- **Headings** : Multi level semantic titles hierarchy .
`<h1> ... <h6>`
- **Paragraphs** : used for standard body blocks of text. `<p>`
- **Links** : created using anchor tags to hyperlink text. `<a href="url">`
- **Lists** : organized structured lines .`<ul> <ol> <li>`



Images and Attributes

The HTML `<img>` tag is used to embed an image in a web page.

```
<img>   Defines an image  
<map>   Defines an image map  
<area>  Defines a clickable area inside an image map  
<picture> Defines a container for multiple image resources
```

- **src Attribute**
specifies the path (URL) to the image.
- **alt Attribute**
provides an alternate text for an image



```

```



Forms : The Interaction Pipeline

Forms convert static markup into interactive user gateways that collect inputs for data handling.

Input Type	HTML Tag	Purpose
Text	<code><input type="text"></code>	Enter text
Password	<code><input type="password"></code>	Enter password
Email	<code><input type="email"></code>	Enter email address
Radio	<code><input type="radio"></code>	Select one option
Checkbox	<code><input type="checkbox"></code>	Select multiple options

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname">
</form>
```



Form Validation & Defense

Ensures users enter correct and complete information before the form is submitted.

required: Makes a field mandatory so it cannot be left empty.

placeholder: Displays a hint or example inside the input field.

minlength & maxlength: Set the minimum and maximum number of characters allowed.

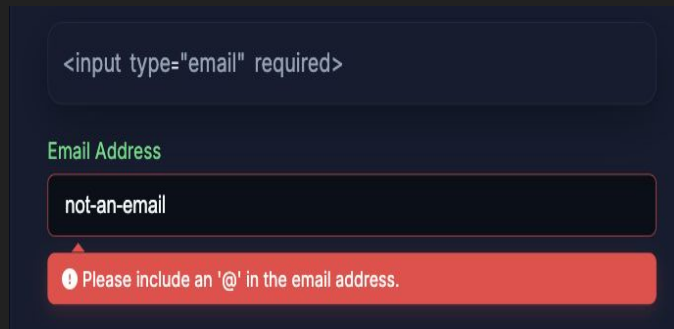
min & max: Restrict the minimum and maximum values for numeric inputs.

pattern: Validates the input using a specific format or regular expression.

readonly: Allows users to view the field but prevents them from editing it.

disabled: Disables the input field so users cannot interact with it.

autocomplete: Allows the browser to suggest previously entered values.



`<input type="email" required>`

Email Address

not-an-email

⚠ Please include an '@' in the email address.



HTML Table

It help organize data into rows and columns, making information easy to read and compare

Table Structure

- <table>** creates the table
- <tr>** defines a table row
- <th>** defines a table header cell
- <td>** Defines a standard table data cell

Company	Contact	Country
Broadway Infosys	97798XXXXXX	Kathmandu



```
<table>
  <tr>
    <th>Company</th>
    <th>Contact</th>
    <th>Country</th>
  </tr>
  <tr>
    <td>Broadway Infosys</td>
    <td>97798XXXXXX</td>
    <td>Kathmandu</td>
  </tr>
</table>
```



Spanning Multiple Rows or Columns

Sometimes you need a cell to stretch across more than one column or down multiple rows (like merging cells in Excel). You can achieve this using two handy attributes:

colspan: Merges columns horizontally.

rowspan: Merges rows vertically.

```
<table border="1">
  <tr>
    <td rowspan="2">Team A</td>
    <td>France</td>
  </tr>

  <tr>
    <td>Germany</td>
  </tr>

  <tr>
    <td colspan="2">Total Country: 2</td>
  </tr>
</table>
```

Team A (Spans 2 Rows Down)	Alice
	Bob
Total Members: 2 (Spans 2 Columns Wide)	



Semantic HTML & Best Practice

It use of HTML tags that clearly define the purpose of the content they enclose to make your website code more understandable for both developers and search engines.

Clear Structuring

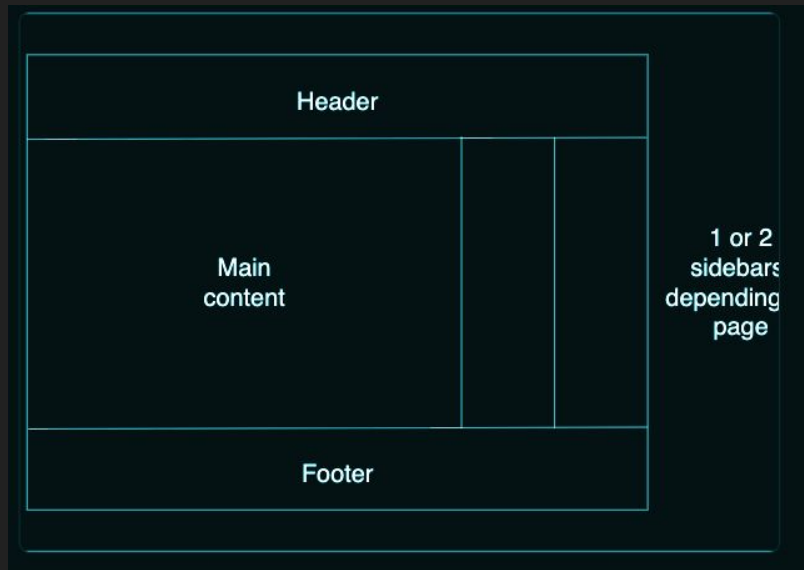
Using designated elements makes your page clean. It eliminates the confusion of reading through dozens of unlabelled, messy lines of code.

Easily Readable Blueprints

Your HTML behaves like a map. A clean layout ensures that web browsers, search engine crawlers, and future developers understand your page instantly.

The Universal Rule

Choose the tag that best describes the content. Never use a generic tag when a built-in meaningful element exists.



Four Vital Page Elements Explained

<header>

Located at the top. Houses your main title, logo, and welcome message for the users.

<nav>

The navigation block. Contains links to help visitors jump to other sections of the page or site.

```
<header>
  <h1>My Website</h1>
</header>

<nav>
  <a href="#">Home</a> |
  <a href="#">About</a> |
  <a href="#">Contact</a>
</nav>

<article>
  <h2>Welcome to My Website</h2>
  <p>This is a simple webpage created using semantic HTML tags.</p>
</article>

<footer>
  <p>&copy; 2026 My Website. All rights reserved.</p>
</footer>
```

<article>

The main topic holder. Used for stand-alone stories, news items, or blog entries.

<footer>

Found at the bottom. Holds copyright info, quick contact details, and secondary links.



SEO Practices

SEO stands for Search Engine Optimization. It means making simple adjustments to your website design so that search engines like Google can find and display your site easily.

- **Use Clear Titles**

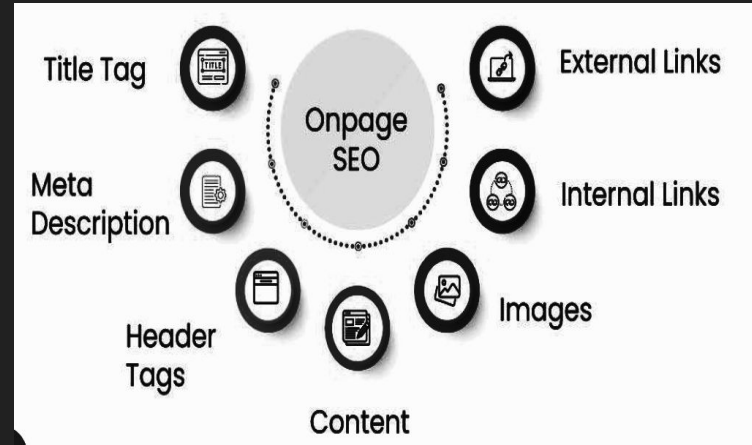
Your heading tags (<h1>, <h2>) should show what the page content is about.

- **Image Alt Descriptions**

Always describe your images so that search engines can "see" what's inside them.

- **Clean Page Structure**

Use logical tags rather than styling generic containers for all content.



Version Control With Git

Imagine writing a long program, making a mistake, and wanting to undo all your work safely. Git keeps snapshots so you can travel back in time!

- **No more duplicate folders:** Say goodbye to files like `final_code_v2_new_v3.html`.
- **Safe testing:** Create temporary code sandboxes without messing up your main project.
- **Collaborative:** Allows multiple team members to code inside the same file at the same time.

```
# Check if Git is installed on your computer

$git --version
git version 2.45.2

# Tell Git who you are (Only needed once!)

$git config --global user.name "Your Name"
$git config --global user.email "your email"
```



Essential Git FLOW

Start Repository

Set up a blank tracking workspace locally, or copy an existing repository from the internet.

```
git init  
git clone [url]
```

Stage Changes

Add specific updated files into the virtual 'packing box' to be saved in the next step.

```
git add index.html
```

Commit Snapshot

Take a permanent local save snapshot accompanied by a short note describing changes.

```
git commit -m  
"msg"
```

Push to Remote

Send your committed snapshots to a cloud repository like GitHub for safekeeping or sharing.

```
git push origin  
main
```



THANK YOU!



Shree Ganesh Marg, Subidhanagar, Tinkune, Kathmandu

01-4117578 / 411849 / 9841002000

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